

QM1

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Abstract

This paper is part of a larger set of papers by Harry Bedford and Samuel Williams, created for the 2026 BPhO (British Physics Olympiad) Computational Challenge. This paper will cover only Quantum Mechanics Problems Set 2, to find sets 1 and 3 along with Nuclear, Particles, and radioactivity Questions, see folder. The range of problems covered here will cover questions surrounding the EM spectrum, Black Body Radiation, The Photoelectric Effect and many more. See Contents for more info.

1 Question 1

1.1 Radiation of caesium radiation

To calculate the wavelength of the radiation, I can use the given frequency. If there are 9,192,631,770 periods of radiation in a second, then,

$$f = 9,192,631,770 \text{ Hz} = 9.1926 \times 10^9 \text{ Hz} \quad (1)$$

This can then be used in the wave-speed equation,

$$\lambda = \frac{c}{f} \quad (2)$$

$$c = 2.998 \times 10^8 \quad (3)$$

$$\lambda = \frac{2.998 \times 10^8}{9.1926 \times 10^9} \simeq 3.26 \times 10^{-2} \text{ m} \quad (4)$$

This is approximately 3.2 cm, which is within the microwave radiation range, which has a wavelength between 0.1cm and 300m.

1.2 Energy change of caesium radiation

To find the energy change, you can use,

$$eV = hf \quad (5)$$

where V is equivalent to the accelerating voltage on one electron, so the energy in electron-volts is,

$$E = \frac{hf}{e} = \frac{6.63 \times 10^{-34} \times 9.1926 \times 10^9}{1.6 \times 10^{-19}} = 3.81 \times 10^{-5} \quad (6)$$

The ionization energy of Hydrogen is 13.6eV, and typical photoelectric effect work functions for metals are 5eV. So this transition is electronic, as nuclear transitions are on the order of MeV.

2 Question 2

2.1 Work function of a metal

Energy of an electron emitted due to absorption of a photon of frequency f is,

$$E = hf - \Phi \quad (7)$$

If stopping potential is V ,

$$eV = hf - \Phi \quad (8)$$

So if its threshold wavelength is 300nm, then its threshold frequency is,

$$f = \frac{c}{\lambda} = \frac{2.998 \times 10^8}{3 \times 10^{-7}} = 9.993 \times 10^{14} \quad (9)$$

Hence, the work function is,

$$\Phi = \frac{2.998 \times 10^8 \times 6.63 \times 10^{-34}}{3 \times 10^{-7} \times 1.6 \times 10^{-19}} = 4.14\text{eV} \quad (10)$$

2.2 current generated from photoelectrons

using the given values of power, 100W, and laser wavelength, 200nm, we can find the energy per laser photon from,

$$hf = \frac{hc}{\lambda} \quad (11)$$

and energy of the photoelectron is,

$$\frac{hc}{\lambda} - \Phi \quad (12)$$

using Φ from 2.1, and if 10% efficient, 10W results in photoelectrons, so,

$$\text{electrons per second} = \frac{10}{\frac{hc}{\lambda} - \Phi} \quad (13)$$

and, so therefore,

$$I = \frac{10e}{\frac{hc}{\lambda} - \Phi} \quad (14)$$

$$I = \frac{10 \times 1.6 \times 10^{-19}}{\frac{6.63 \times 10^{-34} \times 2.998 \times 10^8}{2 \times 10^{-7}} - 4.141 \times 1.6 \times 10^{-19}} \quad (15)$$

$$I = 4.83 \quad (16)$$

3 Question 3

3.1 De-Broglie wavelength of a water molecule

To estimate the De-Broglie wavelength of a water molecule, we must use the definition of the De-Broglie wavelength,

$$\lambda = \frac{h}{p} \quad (17)$$

Now we need the momentum of a water molecule, this can be found by looking at the kinetic energy of it,

$$E_k = \frac{p^2}{2m} = \frac{3}{2} \times k_B T \quad (18)$$

$$p = \sqrt{3k_B T m} \quad (19)$$

Therefore,

$$\lambda = \frac{h}{\sqrt{3k_B T m}} \quad (20)$$

As we are assuming the atomic structure to be $2 \times {}^1_1\text{H} + {}^{16}_8\text{O}$ and that the total mass to be the sum of the neutron and proton parts, ignoring electron mass, the mass of the molecule is,

$$\begin{aligned} m &\simeq 2 \times m_p + 8 \times m_p + 8 \times m_n \\ &\simeq 2 \times 1.6726 \times 10^{-27} + 8 \times 1.6726 \times 10^{-27} + 8 \times 1.6749 \times 10^{-27} \\ &= 3.0125 \times 10^{-26} \end{aligned} \quad (21)$$

So the De-Broglie wavelength is,

$$\lambda = \frac{6.63 \times 10^{-34}}{\sqrt{3 \times 1.38 \times 10^{-23} \times 300 \times 3.0125 \times 10^{-26}}} \quad (22)$$

$$\lambda = 3.43 \times 10^{-11} \text{m} \quad (23)$$

4 Question 4

4.1 Voltage to accelerate an electron to $\frac{c}{2}$

To find the voltage required we need to use the equation

$$eV = \frac{1}{2} m_e v^2 \quad (24)$$

given that we want the electron to reach half the speed of light,

$$\text{let } v = \frac{c}{2} \quad (25)$$

$$eV = \frac{1}{2}m_e \frac{c^2}{4} \quad (26)$$

$$8eV = m_e c^2 \quad (27)$$

$$\begin{aligned} V &= \frac{m_e c^2}{8e} \\ &= \frac{9.109 \times 10^{-31} \times (2.998 \times 10^8)^2}{8 \times 1.60 \times 10^{-19}} \\ &= 6.40 \times 10^4 \text{V} \end{aligned} \quad (28)$$

So, the voltage required to accelerate an electron to half the speed of light is 64kV, at least according to classical physics, you can instead approach this using relativistic physics. To do this we can look at the formula,

$$E = (\gamma - 1)m_e c^2 = eV \quad (29)$$

$$\text{where } \gamma = (1 - \frac{v^2}{c^2})^{-\frac{1}{2}} \quad (30)$$

As we want the electron to reach half the speed of light, we let $\frac{v}{c} = \frac{1}{2}$, so,

$$\gamma = (1 - \frac{1}{4})^{-\frac{1}{2}} = (\frac{3}{4})^{-\frac{1}{2}} = (\frac{4}{3})^{\frac{1}{2}} \quad (31)$$

$$= \frac{2}{\sqrt{3}} \quad (32)$$

So, substituting this value back into the kinetic energy formula,

$$V = (\frac{2}{\sqrt{3}} - 1) \frac{m_e c^2}{e} \quad (33)$$

$$= (\frac{2}{\sqrt{3}} - 1) \times \frac{9.109 \times 10^{-31} \times (2.998 \times 10^8)^2}{1.60 \times 10^{-19}} \quad (34)$$

$$= 7.92 \times 10^4 \text{V} \quad (35)$$

So, the voltage required to accelerate an electron to half the speed of light is 79kV according to relativistic physics.

4.2 De-Broglie wavelength of an electron

To calculate the De-Broglie wavelength of an electron, we can use the De-Broglie equation,

$$\lambda = \frac{h}{p} \quad (36)$$

4.2.1 Classical

For a classical approach, we find momentum from,

$$\frac{p^2}{2m} = eV \quad (37)$$

$$\therefore p = \sqrt{2meV} \quad (38)$$

$$(39)$$

So, substituting back into the De-Broglie equation,

$$\lambda = \frac{h}{\sqrt{2meV}} \quad (40)$$

$$\text{So, using the classical voltage} \quad (41)$$

$$= \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 9.109 \times 10^{-31} \times 1.60 \times 10^{-19} \times 6.40 \times 10^4}} \quad (42)$$

$$= 4.85 \times 10^{-12} \text{m} \quad (43)$$

So, the classical wavelength of an electron is 4.85pm

4.2.2 Relativistic

For a Relativistic approach, we find momentum from,

$$p^2 c^2 + m_e^2 c^4 = (m_e^2 + eV)^2 \quad (44)$$

$$\therefore p = \sqrt{\frac{(m_e^2 + eV)^2 - m_e^2 c^4}{c^2}} \quad (45)$$

substituting, into the De-broglie equation,

$$\lambda = \frac{hc}{\sqrt{(m_e^2 + eV)^2 - m_e^2 c^4}} \quad (46)$$

$$= \frac{hc}{\sqrt{2m_e^2 eV + e^2 V^2}} \quad (47)$$

$$\text{So, using the Relativistic voltage} \quad (48)$$

$$(49)$$

$$\lambda = \frac{6.63 \times 10^{-34} \times 2.998 \times 10^8}{\sqrt{2 \times 9.109 \times 10^{-31} \times (2.998 \times 10^8)^2 \times 1.60 \times 10^{-19} \times 7.92 \times 10^4 + (1.60 \times 10^{-19} \times 7.92 \times 10^4)}} \quad (50)$$

$$\lambda = 4.20 \times 10^{-12} \text{m} \quad (51)$$

So, the classical wavelength of an electron is 4.20pm

4.3 Calculating the lattice spacing

To find the atomic lattice spacing that allows an electron travelling at half the speed of light to be diffracted by 2° , we can use the diffraction grating formula

$$d \sin \theta^\circ = n\lambda \quad (52)$$

So, when $n = 1$, $\theta = 2^\circ$, and $\lambda = 4.20 \times 10^{-12}$ for the relativistic situation,

$$d \sin 2^\circ = 1 \times 4.20 \times 10^{-12} \quad (53)$$

$$d = \frac{4.20 \times 10^{-12}}{\sin 2^\circ} \quad (54)$$

$$\simeq 1.2 \times 10^{-10} \text{m} \quad (55)$$

So, for an electron to be diffracted 2° at half the speed of light, the atomic lattice spacing needs to approximately be $1.2\text{\AA}$

5 Question 5

5.1 Greatest wavelength to ionize

To find the greatest wavelength of EMR that will ionize an atom, we need to use bohr model,

$$E_n = -13.6\text{eV} \times \frac{Z^2}{n^2} \quad (56)$$

So, when we relate this with the Planck-Einstein relation assuming $n=1$, we end up with,

$$E = hf = \frac{hc}{\lambda} \quad (57)$$

$$\lambda = \frac{hc}{13.6eZ^2} \quad (58)$$

5.1.1 Hydrogen

So, for Hydrogen with atomic number 1, we can substitute this in and find,

$$\lambda_H = \frac{6.63 \times 10^{-34} \times 2.998 \times 10^8}{13.6 \times 1.60 \times 10^{-19} \times 1} \quad (59)$$

$$= 91.3\text{nm} \quad (60)$$

5.1.2 Helium

As Helium has atomic number 2, its wavelength will be $\frac{\lambda_H}{2^2}$ so,

$$\lambda_{\text{He}} = \frac{91.3}{2^2} \quad (61)$$

$$= 22.8\text{nm} \quad (62)$$

5.1.3 Radon

We can do a similar process as we did with helium with radon,

$$\lambda_{\text{Rn}} = \frac{91.3}{86^2} \quad (63)$$

$$= 0.0124\text{nm} \quad (64)$$

5.2 Comparison of experimental and calculated values

To calculate the ionization energy for each element according to the bohr model we can use

$$E = 13.6\text{eV} \times Z^2 \quad (65)$$

So using this for each element,

$$E_H = 13.6\text{eV} \quad (66)$$

$$E_{\text{He}} = 13.6\text{eV} \times 2^2 = 54.4\text{eV} \quad E_{\text{Rn}} = 13.6\text{eV} \times 86^2 = 100,585.6\text{eV} \quad (67)$$

However, all these values are in electron-volts, while the given experimental values are in kJmol^{-1} , so to be able to comment on the difference between these values we need to convert the experimental values in the same units as the calculated ones. To do this we need to use the relationship

$$E_{\text{eV}} = \frac{E_{\text{Jmol}^{-1}}}{N_A e} \text{ where } N_A \text{ is Avogadros number} \quad (68)$$

$$\therefore E_{\text{eV}} = \frac{E_{\text{kJmol}^{-1}}}{N_A e \times 10^{-3}} \quad (69)$$

So, using the given experimental values, we get

$$E_H = \frac{1312}{6.0221 \times 10^{23} \times 1.60 \times 10^{-19} \times 10^{-3}} \quad (70)$$

$$\simeq 13.6\text{eV} \quad (71)$$

$$E_{\text{He}} = \frac{2372.3}{6.0221 \times 10^{23} \times 1.60 \times 10^{-19} \times 10^{-3}} \quad (72)$$

$$\simeq 24.6\text{eV} \quad (73)$$

$$E_{\text{Rn}} = \frac{1037}{6.0221 \times 10^{23} \times 1.60 \times 10^{-19} \times 10^{-3}} \quad (74)$$

$$\simeq 10.8\text{eV} \quad (75)$$

As, you can see above, the Bohr model is very good for low values of Z, as Hydrogen is correct to one decimal place, however as Z increases it becomes worse and worse, even at Z=2 for Helium, the expected value according to Bohr is almost double the experimental. For Radon, it is out by a factor of 10^4 , this is due to that the model of one electron exposed to Ze charge is incorrect for larger atoms. The repulsion from other electrons must be significant, especially as Z increases.

5.3 Actual wavelength of photons for these energies

To find the wavelengths associated with the experimental energy values we can use the Planck-Einstein relation,

$$\lambda = \frac{hc}{E} \quad (76)$$

However, as our energy is in eV, we can use the shortcut

$$\lambda = \frac{1242\text{nm}}{E_{\text{eV}}} \quad (77)$$

So, for the given energies,

$$\lambda_H = \frac{1242}{13.6} = 91.3\text{nm} \quad (78)$$

$$\lambda_{\text{He}} = \frac{1242}{24.6} = 50.5\text{nm} \quad (79)$$

$$\lambda_{\text{Rn}} = \frac{1242}{10.8} = 115.0\text{nm} \quad (80)$$

$$(81)$$

6 Question 6

6.1 Number of photons received per second by FAST

If we assume that the Andromeda nebula radiates isotopically at 8×10^{27} W at 1420 MHz, then the power per unit area distance, d , away is

$$\frac{P}{4\pi d^2} \quad (82)$$

So, the power received by FAST, which is modeled as a circle of radius R ,

$$\pi R^2 \times \frac{P}{4\pi d^2} = \frac{1}{4} P \left(\frac{R}{d} \right)^2 \quad (83)$$

Given that each photon has energy hf , we can find the number of photons received by FAST per second can be found by,

$$= \frac{\text{Energy per second}}{\text{photon energy}} \quad (84)$$

$$= \frac{\frac{1}{4} P \left(\frac{R}{d} \right)^2}{hf} \quad (85)$$

$$= \frac{PR^2}{4d^2 hf} \quad (86)$$

So, using our assumed values for FAST, with $R = 250$ m,

$$= \frac{8 \times 10^{27} \times 250^2}{4 \times (2 \times 10^{22})^2 \times 6.63 \times 10^{-34} \times 1420 \times 10^6} = 3.3 \times 10^{11} \text{ photons/s} \quad (87)$$

6.2 Comparison with a DSLR camera

To find the amount of energy a DSLR receives,

$$E = \text{Area of CCD} \times \text{Power of input light} \times \text{shutter speed} \quad (88)$$

So, when substituting in given values, we get the following

$$E = (24 \times 10^{-3} \times 36 \times 10^{-3}) \times 100 \times \frac{1}{500} \quad (89)$$

$$= 1.728 \times 10^{14} \text{ Js}^{-1} \quad (90)$$

So, using the equation previously found to find the number of photons, assuming that the wavelength of sunlight is 500 nm,

$$= \frac{\text{Energy per second}}{\text{photon energy}} \quad (91)$$

$$= \frac{1.728 \times 10^{-4}}{h \times \frac{c}{\lambda}} \quad (92)$$

$$= \frac{1.728 \times 10^{-4}}{6.63 \times 10^{-34} \times \frac{2.998 \times 10^8}{500 \times 10^{-9}}} \quad (93)$$

$$= 4.35 \times 10^{14} \text{ photons/s} \quad (94)$$

Therefore, to take an exposure similar to that of a DSLR camera with FAST, it would need to stare at Andromeda for $\frac{4.35 \times 10^{14}}{3.3 \times 10^{11}}$ seconds or $\simeq 1318 \text{ s} \simeq 22$ minutes

7 Question 7

7.1 Calculating impulse on the solar sail

using the De-Broglie relation, with wavelength λ allows us to find

$$\lambda = \frac{h}{p} \quad (95)$$

$$p = \frac{h}{\lambda} \quad (96)$$

Now once photon has reflected, its direction is reversed, so the sail must provide an impulse of $2p$. So each photon exerts an impulse of $2p$ on the sail. So, an

equation for impulse can be found from

$$\Delta p = 2p \quad (97)$$

$$\Delta p = \frac{2h}{\lambda} \quad (98)$$

7.2 Calculating acceleration of a solar sail propelled spaceship

The energy of each photon can be found by using Planck's relation,

$$E = hf = \frac{hc}{\lambda} \quad (99)$$

Hence, the number of photons striking the sail per second, where I is intensity, and A is area of the solar sail, can be found from

$$\frac{I \times A}{\frac{hc}{\lambda}} \quad (100)$$

So, the total impulse per second (which is equal to the force upon the sail) can be found from

$$F = \text{photons per second} \times \text{impulse on sail per photon} \quad (101)$$

$$= \frac{IA\lambda}{hc} \times \frac{2h}{\lambda} \quad (102)$$

$$F = \frac{2IA}{c} \quad (103)$$

Now, to find the acceleration of a the spaceship, we can use Newtons II law

$$F = ma \quad (104)$$

$$a = \frac{F}{m} \quad (105)$$

$$a = \frac{2IA}{mc} \quad (106)$$

7.3 Calculating the time taken to accelerate to $\frac{c}{100}$

Assuming light has an intensity of 1 kWm^{-2} , and the mass of the spacecraft is 10^4 kg , with a solar sail of area 10^8 m^2 . When we apply this using classical

physics, we can find the time to reach $\frac{c}{100}ms^{-1}$

$$v = at \quad (107)$$

$$\therefore t = \frac{v}{a} \quad (108)$$

$$\therefore t = v \times \frac{mc}{2IA} \quad (109)$$

$$\text{So, if } v = \frac{c}{100}, \quad (110)$$

$$t = \frac{mc^2}{200IA} \quad (111)$$

$$= \frac{10^4 \times (2.998 \times 10^8)^2}{200 \times 10^3 \times 10^8} \quad (112)$$

$$= 4.49 \times 10^7 \text{seconds} \quad (113)$$

$$\simeq 1 \text{ year, 5 months} \quad (114)$$

8 Question 8

8.1 Estimating the total power emitted by the Sun

International Space Station receives 1361 Wm^{-1} from the Sun. If the Sun radiates isotropically, and $L_{\odot}$ is the total power emitted, then

$$L_{\odot} = 1361 \times 4\pi \times \text{AU}^2 \quad (115)$$

$$= 1361 \times 4\pi \times (1.496 \times 10^{11})^2 \quad (116)$$

$$L_{\odot} = 3.83 \times 10^{26} \text{W} \quad (117)$$

If the sun shines for 10 billion years, this means energy radiated is,

$$E = 3.83 \times 10^{26} \times 10^{10} \times 365 \times 24 \times 3600 \quad (118)$$

$$E = 1.21 \times 10^{44} \text{J} \quad (119)$$

Now, if we use the mass-energy equivalence relation, we can find how much mass the sun would lose over it's time spent in main phase,

$$E = \Delta mc^2 \quad (120)$$

$$\Delta m = \frac{E}{c^2} \quad (121)$$

$$= \frac{1.21 \times 10^{44}}{(2.998 \times 10^8)^2} \quad (122)$$

$$= 1.34 \times 10^{27} \quad (123)$$

$$\text{Given that} \quad (124)$$

$$M_{\odot} = 1.989 \times 10^{30} \quad (125)$$

$$\therefore \%change = \frac{1.34 \times 10^{27}}{1.989 \times 10^{30}} \times 100 \simeq 0.07\% \quad (126)$$

8.2 Questions about Sirius

8.2.1 Earth orbiting Sirius

We can Calculate the maximum power that would be received on earth by comparing the intensities of Sirius and the Sun

$$I = \sigma T^4 \quad (127)$$

$$\therefore I \propto T^4 \quad (128)$$

$$\frac{I_{Sirius}}{I_{Sun}} = \left(\frac{9940}{5778} \right)^4 = 8,759 \quad (129)$$

$$(130)$$

Now, when this value is multiplied by the normal maximum power received on Earth (1,361 as per last part) we get the maximum that would be received on Earth from Sirius.

$$1361 \times 8759 = 11.9 kW m^{-2} \quad (131)$$

8.2.2 Luminosity of Sirius

$$L_{Sirius} = \left(\frac{9940}{5778} \right)^4 \times L_{\odot} \times \left(\frac{R_{Sirius}}{R_{Sun}} \right)^2 \quad (132)$$

$$= \left(\frac{9940}{5778} \right)^4 \times 3.83 \times 10^{26} \times \left(\frac{1.711}{1} \right)^2 \quad (133)$$

$$= 9.82 \times 10^{27} J s^{-1} \quad (134)$$

8.2.3 Estimating Sirius' time in main phase

If Sirius loses 0.07% of it's mass over it's time, t, in main phase, then we can use the mass-energy equivalence relation

$$E = mc^2 \quad (135)$$

$$\text{So to find time} \quad (136)$$

$$Et = \frac{\text{percentage of mass lost}}{100} \times mc^2 \quad (137)$$

$$= \frac{\text{percentage of mass lost}}{100} \times 2.02 \times m_{\odot} \times c^2 \quad (138)$$

$$9.82 \times 10^{27} t = \frac{0.07}{100} \times 2.02 \times 1.989 \times 10^{30} \times (2.998 \times 10^8)^2 \quad (139)$$

$$\therefore t = 2.57 \times 10^{16} s \quad (140)$$

$$\simeq 0.82 \text{ billion years} \quad (141)$$

8.3 Time for radiation from Sirius to boil water

To find the time for radiation to boil water, first we must find the power received by a 10 m² solar panel at 1 AU from Sirius, which can be found from

$$P = AI \quad (142)$$

$$= 10 \times \frac{9.82 \times 10^{27}}{4\pi \times (1.496 \times 10^{11})^2} \quad (143)$$

$$P = 349 \text{ kW} \quad (144)$$

One litre (1 kg) of water at 20°C requires energy, E, to boil

$$E = mc\Delta T \quad (145)$$

$$= 4181 \times 80 \quad (146)$$

$$E = 3.345 \times 10^5 J = 334.5 kJ \quad (147)$$

So, the solar panel will boil the water in time, t,

$$t = \frac{\text{energy required}}{\text{energy output/s}} \quad (148)$$

$$= \frac{334.5}{349} \quad (149)$$

$$= 0.96 \text{ s} \quad (150)$$